

# SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Dept. Name:CSE -AIDS

Course Code/Course Name: DSA06 /Data Handling and Visualization

## LIST OF EXPERIMENTS

### Set 1: Monthly Sales Data

#### Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.
2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.
3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

#### SOLUTIONS:

### 1. Create the dataset

# Monthly Sales Dataset

```
sales_data <- data.frame(  
  Month = factor(month.name, levels = month.name),  
  Sales = c(45000, 48000, 52000, 50000, 58000, 62000,  
            60000, 65000, 70000, 68000, 75000, 80000),  
  Advertising = c(5000, 5500, 6000, 5800, 6500, 7000,  
                 6800, 7500, 8000, 7800, 8500, 9000)  
)  
# Display the data  
print(sales_data)
```

### 2. Line Chart – Monthly Sales

# Line Chart

```
plot(sales_data$Month,  
     sales_data$Sales,
```

```
type = "o",  
col = "blue",  
pch = 16,  
xlab = "Month",  
ylab = "Sales",  
main = "Monthly Sales",  
las = 2)
```

### **3. Bar Chart – Top-Selling Products**

For this question, we need product-wise sales data.

```
# Product Sales Data  
product_data <- data.frame(  
  Product = c("Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch"),  
  Sales = c(950000, 1200000, 750000, 450000, 600000)  
)  
# Sort products by sales  
product_data <- product_data[order(product_data$Sales, decreasing = TRUE), ]  
# Bar Chart  
barplot(product_data$Sales,  
  names.arg = product_data$Product,  
  col = "steelblue",  
  main = "Top-Selling Products for the Year",  
  xlab = "Products",  
  ylab = "Annual Sales",  
  las = 2)
```

This displays the products in descending order of annual sales.

### **4. Scatter Plot – Advertising Budget vs Monthly Sales**

```
# Scatter Plot  
plot(sales_data$Advertising,  
  sales_data$Sales,  
  pch = 19,  
  col = "darkgreen",  
  xlab = "Advertising Budget",  
  ylab = "Monthly Sales",
```

```

    main = "Advertising Budget vs Monthly Sales")
# Add trend line
abline(lm(Sales ~ Advertising, data = sales_data),
      col = "red",
      lwd = 2)

```

## Set 2: Customer Feedback Analysis

### Dataset: Customer Satisfaction

**Customer ID Age Satisfaction Score**

1 25 4

2 30 5

3 35 3

4 28 4

5 40 5

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.
4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

### SOLUTIONS:

#### 1. Histogram – Distribution of Customer Ages

```
# Histogram of Customer Ages
```

```

hist(customer_data$Age,
     breaks = 5,
     col = "skyblue",
     border = "black",
     main = "Distribution of Customer Ages",
     xlab = "Age",
     ylab = "Number of Customers")

```

## 2. Pie Chart – Satisfaction Score Distribution

```
# Calculate frequency of each satisfaction score
satisfaction_count <- table(customer_data$Satisfaction_Score)
# Pie Chart
pie(satisfaction_count,
    labels = paste(names(satisfaction_count),
                    "Score (", satisfaction_count, ")", sep = ""),
    main = "Customer Satisfaction Score Distribution",
    col = rainbow(length(satisfaction_count)))
```

## 3. Stacked Bar Chart – Satisfaction by Age Group

First, we create age groups.

```
# Create Age Groups
customer_data$Age_Group <- cut(
  customer_data$Age,
  breaks = c(20, 30, 40, 50),
  labels = c("21-30", "31-40", "41-50"),
  include.lowest = TRUE
)
```

```
# Display updated data
print(customer_data)
```

Now create the stacked bar chart:

```
# Create table of Age Group and Satisfaction Score
```

```
age_satisfaction <- table(
  customer_data$Age_Group,
  customer_data$Satisfaction_Score
)
# Stacked Bar Chart
barplot(age_satisfaction,
  beside = FALSE,
  col = c("lightblue", "lightgreen", "orange"),
  main = "Satisfaction Scores by Age Group",
```

```
xlab = "Age Group",  
ylab = "Number of Customers",  
legend.text = TRUE)
```

#### **4. Word Cloud – Customer Feedback**

```
# Open-ended customer feedback  
feedback <- c(  
  "Excellent service and very friendly staff",  
  "Very satisfied with the product and service",  
  "Good quality and fast service",  
  "The product was excellent and easy to use",  
  "Friendly staff and excellent customer support"  
)
```

#### **Generate Word Cloud**

```
# Create Corpus  
corpus <- Corpus(VectorSource(feedback))  
# Convert text to lowercase  
corpus <- tm_map(corpus, content_transformer(tolower))  
# Remove punctuation  
corpus <- tm_map(corpus, removePunctuation)  
# Remove numbers  
corpus <- tm_map(corpus, removeNumbers)  
# Remove common stop words  
corpus <- tm_map(corpus, removeWords, stopwords("english"))  
# Create Word Cloud  
wordcloud(corpus,  
  min.freq = 1,  
  max.words = 50,  
  random.order = FALSE,  
  colors = rainbow(8),  
  scale = c(3, 0.5),  
  main = "Customer Feedback Word Cloud")
```

### Set 3: Employee Performance Evaluation

Month Sales (in \$)

January 15000

February 18000

March 22000

April 20000

May 23000

Dataset: Employee Performance

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

3. Build a scatter plot to analyze the correlation between years of service and performance scores.

Explain any insights.

4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

### SOLUTIONS:

#### 1. Line Chart – Performance Trend Over Time

# Line Chart - Monthly Performance/Sales Trend

```
plot(monthly_sales$Month,  
     monthly_sales$Sales,  
     type = "o",  
     pch = 16,  
     col = "blue",  
     xlab = "Month",  
     ylab = "Sales ($)",  
     main = "Monthly Performance Trend",  
     las = 2)
```

# Add legend

```
legend("topleft",  
     legend = "Monthly Sales",  
     col = "blue",  
     lty = 1,  
     pch = 16)
```

## 2. Bar Chart – Employees by Department

```
# Count employees in each department
department_count <- table(employee_data$Department)
# Bar Chart
barplot(department_count,
        col = c("steelblue", "orange", "green"),
        main = "Distribution of Employees by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        ylim = c(0, 3))
# Add values above bars
text(x = seq_along(department_count),
     y = department_count,
     labels = department_count,
     pos = 3)
```

## 3. Scatter Plot – Years of Service vs Performance

```
# Scatter Plot
plot(employee_data$Years_of_Service,
     employee_data$Performance_Score,
     pch = 19,
     col = "darkgreen",
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Years of Service vs Performance Score")
# Add employee IDs as labels
text(employee_data$Years_of_Service,
     employee_data$Performance_Score,
     labels = employee_data$Employee_ID,
     pos = 3)
# Add trend line
model <- lm(Performance_Score ~ Years_of_Service,
           data = employee_data)
abline(model,
       col = "red",
       lwd = 2)
```

## Set 4: Product Inventory Management

### Dataset: Product Inventory

Product ID Product Name Quantity Available

1 Product A 250

2 Product B 175

3 Product C 300

4 Product D 200

5 Product E 220

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title

the chart.

2. Generate a stacked bar chart to show the quantity of each product within different product categories.

3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

### SOLUTIONS:

#### 1. Bar Chart – Quantity of Each Product

# Bar Chart

```
barplot(inventory_data$Quantity_Available,  
        names.arg = inventory_data$Product_Name,  
        col = "steelblue",  
        main = "Product Inventory Quantity",  
        xlab = "Product",  
        ylab = "Quantity Available",  
        ylim = c(0, 350))
```

# Add quantity values above bars

```
text(x = seq_along(inventory_data$Quantity_Available),  
     y = inventory_data$Quantity_Available,  
     labels = inventory_data$Quantity_Available,  
     pos = 3)
```

## 2. Stacked Bar Chart – Products by Category

```
# Add Product Categories
inventory_data$Category <- c(
  "Electronics",
  "Electronics",
  "Furniture",
  "Furniture",
  "Accessories"
)
# Display updated dataset
print(inventory_data)
```

Now create the stacked bar chart:

```
# Create category-wise quantity table
category_quantity <- xtabs(
  Quantity_Available ~ Category + Product_Name,
  data = inventory_data
)
# Stacked Bar Chart
barplot(category_quantity,
  beside = FALSE,
  col = c("skyblue", "orange", "lightgreen"),
  main = "Product Quantity by Category",
  xlab = "Product",
  ylab = "Quantity Available",
  legend.text = rownames(category_quantity))
```

## 3. Scatter Plot – Product Price vs Quantity Available

```
# Add Product Prices
inventory_data$Price <- c(
  500,
  750,
  1200,
  900,
```

```

650
)
# Display updated dataset
print(inventory_data)

Create Scatter Plot
# Scatter Plot
plot(inventory_data$Price,
     inventory_data$Quantity_Available,
     pch = 19,
     col = "darkgreen",
     xlab = "Product Price ($)",
     ylab = "Quantity Available",
     main = "Product Price vs Quantity Available")
# Add product names
text(inventory_data$Price,
     inventory_data$Quantity_Available,
     labels = inventory_data$Product_Name,
     pos = 3)
# Add trend line
model <- lm(
  Quantity_Available ~ Price,
  data = inventory_data
)
abline(model,
       col = "red",
       lwd = 2)

```

## **Set 5: Website Analytics**

### **Dataset: Website Traffic**

#### **Date Page Views Click-through Rate**

2023-01-01 1500 2.3%

2023-01-02 1600 2.7%

2023-01-03 1400 2.0%

2023-01-04 1650 2.4%

Date Page Views Click-through Rate

2023-01-05 1800 2.6%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

### **SOLUTIONS:**

#### 1. Line Chart – Daily Page Views

# Line Chart

```
plot(website_data$Date,  
     website_data$Page_Views,  
     type = "o",  
     pch = 16,  
     col = "blue",  
     xlab = "Date",  
     ylab = "Page Views",  
     main = "Daily Page Views Trend")
```

#### 2. Bar Chart – Top N Days by Click-Through Rate

# Number of top days

```
N <- 3
```

# Sort data by Click-through Rate

```
sorted_data <- website_data[  
  order(-website_data$Click_Through_Rate),  
]
```

# Select top N days

```
top_days <- head(sorted_data, N)  
print(top_days)
```

Create Bar Chart

```

# Bar Chart
barplot(top_days$Click_Through_Rate,
        names.arg = format(top_days$Date,
                            "%Y-%m-%d"),
        col = "orange",
        main = "Top 3 Days by Click-Through Rate",
        xlab = "Date",
        ylab = "Click-Through Rate (%)",
        ylim = c(0, 3))
# Add values above bars
text(x = seq_along(top_days$Click_Through_Rate),
     y = top_days$Click_Through_Rate,
     labels = paste0(
       top_days$Click_Through_Rate,
       "%"),
     pos = 3)

```

### 3. Stacked Area Chart – User Interactions

```

# User Interaction Data
interaction_data <- data.frame(
  Date = as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03",
                  "2023-01-04",
                  "2023-01-05")),
  Likes = c(120, 150, 110, 160, 190),
  Shares = c(50, 65, 45, 70, 80),
  Comments = c(30, 40, 25, 35, 50)
)
print(interaction_data)

```

Stacked Area Chart using Base R

```

# Create matrix for area chart

```

```

interaction_matrix <- t(
  as.matrix(
    interaction_data[, c(
      "Likes",
      "Shares",
      "Comments"
    )]
  )
)
# Stacked Area Chart
area_matrix <- interaction_matrix
# Cumulative values
cumulative <- apply(
  area_matrix,
  2,
  cumsum
)
# Plot first area
plot(interaction_data$Date,
     cumulative[3, ],
     type = "n",
     ylim = c(0, max(cumulative)),
     xlab = "Date",
     ylab = "Number of Interactions",
     main = "User Interactions Over Time")
# Draw stacked areas
polygon(
  c(interaction_data$Date,
    rev(interaction_data$Date)),
  c(cumulative[1, ],
    rev(rep(0, nrow(interaction_data)))),
  col = "lightblue",
  border = "white"
)
polygon(
  c(interaction_data$Date,
    rev(interaction_data$Date)),

```

```

    c(cumulative[2, ],
      rev(cumulative[1, ])),
    col = "lightgreen",
    border = "white"
  )
  polygon(
    c(interaction_data$Date,
      rev(interaction_data$Date)),
    c(cumulative[3, ],
      rev(cumulative[2, ])),
    col = "orange",
    border = "white"
  )
# Add legend
legend(
  "topleft",
  legend = c(
    "Likes",
    "Shares",
    "Comments"
  ),
  fill = c(
    "lightblue",
    "lightgreen",
    "orange"
  )
)
)

```